

Mahmoud Sajjadi

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Research Interests

AI Researcher & Machine Learning Engineer (Ph.D.) expert in distributed learning, foundation models (NLP/LLMs), and mathematical optimization applied to computer vision, computational biology, and clinical diagnostics. Proven R&D track record in multimodal data fusion for tracking early-stage brain cognitive decline (Alzheimer's), deploying scalable, privacy-preserving Edge AI. Proficient in Python and statistical modeling for large-scale biomedical networks and data analytics.

Education

University of Nevada, Reno Ph.D. in Computer Science and Engineering Dissertation: <i>Distributed Machine Learning in Heterogeneous Edge Networks</i>	2022 – 2025
University of Tehran, Iran M.Sc. in Electrical Engineering-Control Systems Thesis: <i>Simultaneous State Estimation and Reinforcement Learning in Stochastic Matrix Games</i>	2011 - 2014
Shiraz University of Technology, Iran B.Sc. in Electrical Engineering-Control Systems	2007 - 2011

Research Experience and Publications

Publications:

- J. Zhang, L. Yang, **S. M. Sajjadi Mohammadabadi**, and F. Yan, "A survey on self-supervised learning: Recent advances and open problems," in *Neurocomputing*, 2025, p. 131409 (Impact Factor: 6.5).
- S. M. Sajjadi Mohammadabadi**, A. K. Moghaddam, M. Entezami, M. Seyedrezaei, D. Charkhian, B. Moghaddam, and M. Sasani, "Advancing Electric Vehicle Ecosystems: A Survey of Generative Artificial Intelligence and Distributed Machine Learning Applications," in *Global Energy Interconnection*, Oct. 2025
- S. M. Sajjadi Mohammadabadi**, B. C. Kara, C. Eyupoglu, C. Uzay, M. S. Tosun, and O. Karakuş, "A Survey of Large Language Models: Evolution, Architectures, Adaptation, Benchmarking, Applications, Challenges, and Societal Implications," in *Electronics*, vol. 14, no. 18, p. 3580, 2025.
- S. M. Sajjadi Mohammadabadi**, L. Yang, F. Yan, and J. Zhang, "Communication-Efficient Training Workload Balancing for Decentralized Multi-Agent Learning" in *IEEE International Conference on Distributed Computing Systems (ICDCS)*, Jul. 2024 (Acceptance rate: 21.9%).
- S. M. Sajjadi Mohammadabadi**, S. Zawad, F. Yan, and L. Yang, "Speed Up Federated Learning in Heterogeneous Environments: A Dynamic Tiering Approach" in *IEEE Internet of Things Journal*, 2024 (Impact Factor: 9.0).
- S. M. Sajjadi Mohammadabadi**, and M. B. Peikani, "Identification and Classification of Rheumatoid Arthritis Using Artificial Intelligence and Machine Learning," in *Diagnosing Musculoskeletal Conditions Using Artificial Intelligence and Machine Learning to Aid Interpretation of Clinical Imaging*, Academic Press, 2025, pp. 123-145.
- S. M. Sajjadi Mohammadabadi**, M. Entezami, A.K. Moghaddam, M. Orangian, and S. Nejadshamsi, "Generative Artificial Intelligence for Distributed Learning to Enhance Smart Grid Communication" in *International Journal of Intelligent Networks*, 2024.
- S. M. Sajjadi Mohammadabadi**, F. Seyedkhamoushi, M. Mostafavi, and M. B. Peikani, "Examination of AI's Role in Diagnosis, Treatment, and Patient Care," chapter in *Transforming Gender-Based Healthcare with AI and Machine Learning*, vol. 1, CRC Press, 2024, pp. 221-238.
- A. Ghasemkhani, R. Sanjeev Haridas, **S. M. Sajjadi Mohammadabadi**, and L. Yang, "Feature collusion attack on PMU data-driven event classification," in *2024 IEEE PES Innovative Smart Grid Technologies (ISGT)*, Washington DC, USA, Feb. 2024.
- S. M. Sajjadi Mohammadabadi**, Y. Liu, A. Canafe, and L. Yang, "Towards distributed learning of PMU data: A federated learning based event classification approach," in *2023 IEEE PES General Meeting*, Orlando, FL, USA, Jul. 2023.

S. M. Sajjadi Mohammadabadi, M. Nili, and H. Kebriaei, “Simultaneous Learning and State Estimation in Repeated Matrix Games,” in 4th International Conference on Information Technology Management, Communication and Computer, (pp. 298- 301), Tehran, Iran, 2014.

Under Review:

SOLAR: Subspace-Oriented Lightweight Adapter Reparameterization for Scalable PEFT Compression, 2026.
A Survey on Hallucination in Large Language Models: Definitions, Detection, and Mitigation 2026

Teaching Experience

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| • Teaching Assistant, University of Nevada Reno, NV | 2023 - present |
| Assisted in teaching core undergraduate & graduate courses Introduction to Big Data, Introduction to Data Mining. | |
| • Sematec, Iran | 2017 – 2022 |
| CDCP (Certified Data Centre Professional) & CDCS (Certified Data Centre Specialist) | |
| • Sematec, Iran | 2018 – 2021 |
| Data Center Standards & Best Practices (TIA-942, BICSI 002-2019, BICSI 009-2019) | |
| • Sematec, Iran | 2019 – 2020 |
| CDCE (Certified Data Centre Expert), | |
| • Teaching Assistant, University of Tehran, Iran | 2011 – 2014 |
| Modern Control | |

Professional Experience

Connected Future Labs, Reno, NV

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| <i>Research and Development Scientist</i> | 2024 – present |
| • Engineered automated multi-modal pipelines to ingest, clean, and harmonize high-frequency streaming data (PPG, EDA, ACC, skin temperature, Muse EEG, and IMU) from wearable Empatica, Muse headsets, and IMU data sensors. | |
| • Extracted physiological biomarkers, e.g., HRV, sympathetic/parasympathetic responses, and neural features, analyzing biometric correlations to map signatures differentiating Mild Cognitive Impairment (MCI) from healthy cohorts. | |
| • Architected AWS S3 and SageMaker pipelines to autonomously audit multi-sensor data collection quality, engineer validation routines for signal anomalies, and drive end-to-end pipelines optimized for ML classification and prediction. | |
| • Led research and development (R&D) of AI-driven methods for monitoring and prediction of early-stage cognitive decline (Alzheimer) using wearable data, advancing early detection capabilities in healthcare applications. | |
| • Researched and analyzed state-of-the-art methods for the detection of Alzheimer's and its measured impact on the human body. | |

Computer Science and Engineering Department, University of Nevada, Reno, NV

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| <i>Graduate Research Assistant</i> | 2022 – Present |
| • Designed and fine-tuned Transformer-based vision and language models (e.g., ViT, LLaMA, GPT-2) using subspace reparameterization, reducing communication and storage by 35% compared to LoRA, enabling on-device deployment (Under review for publication). | |
| • Built and optimized dynamic ML systems using federated learning and split learning for heterogeneous clients, cutting training time by 80% in computer vision applications (Published in IoT journal). | |
| • Developed and deployed decentralized ML using dynamic tiering, reducing slower-side compute by 26% for deep learning models; with convex / non-convex convergence analysis (Presented in ICDCS conference). | |
| • Enhanced federated learning using distance correlation and patch shuffling in SplitFed (Published in the IoT Journal). | |
| • Engineered and implemented ML algorithms with XGBoost and MLP classifiers for classifying current, voltage, and frequency faults in electrical systems, achieving 91% accuracy via federated learning (Presented in PESGM conference). | |
| • Engineered GNN-based predictive models for open source COVID data analysis, achieving 71% accuracy on 181M-node interaction graphs with optimized multiprocessing and graph-based data structures. | |

Battle Born AI, Reno, NV

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| <i>Machine Learning Scientist</i> | 2025 – 2025 |
| • Researched and analyzed LLMs to explore and implement clinical decision-support and business-focused applications. | |

Prochista, Tehran, Iran

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| <i>Electrical Infrastructure Designer, Lecturer, Researcher</i> | 2016 – 2022 |
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- Author of three chapters in data center facility preventive maintenance instructions for electrical systems.
- Designed, managed, maintained, and troubleshoot power, electrical, and cooling systems, as well as Data Center Infrastructure Management (DCIM) and Building Management Systems (BMS).
- Managed and engineered resource utilization monitoring (space and electrical consumption) to improve resource prediction. Enhanced safety measures, including fire alarm integration with the BMS.
- Supervised BMS / DCIM system implementation and maintenance.

Electrical Infrastructure Designer

2014 – 2016

- Author of RFP, request for proposal, detailing technical specifications based on TIA-942 standards for new commercial data center projects.
- Designer of safety and DCIM solutions to ensure end-to-end compliance with the EN 50600 standards.
- Technical reviewer and evaluator of electrical, BMS and fire extinguisher systems.
- Electrical systems monitoring team leader and supervisor for data center expansion and construction projects.

Arya Heavy Machinery - Caterpillar, Tehran, Iran

Power-Electrical Specialist Engineer

2013 – 2014

- Supervisor of the customer training department of the company
- Technical support of the sales team
- Performing market analysis on power-electrical system segment, diesel generators

Professional Certifications, Awards and Honors

- GSA Outstanding Graduate Student Advocacy Award, UNR, 2025
- GSA Outstanding Graduate Researcher Award, UNR, 2024
- Awarded the TechWise program scholarship by TalentSprint Inc., supported by Google (2023 - 2024)
- Python for Data Science and AI by IBM (Coursera 2020)
- Deep Learning with Python and PyTorch by IBM and Deep Learning with TensorFlow by IBM (edX 2020)
- Certified Data Centre Specialist, Professional & Expert (CDCS, CDCP, & CDCE), epi, Iran & Malaysia 2016-2017
- DC102 (Applied Data Center Design and Best Practices | BICSI), TUV Academy, 2016
- Student of National Organization for Development of Exceptional Talents (2000-2007)
- ITIL, Linux, EN50600 standard, Cooling Infrastructure Designing, CDCD (Certified Data Center Designer)

Professional Activities and Service

Graduate Research Assistant – NSF REU Site: Cross-disciplinary Research Experience for Undergraduates on Big Data Analytics in Smart Cities, University of Nevada, Reno 2022–2023

- Contributed to an NSF-funded initiative (Award #1950485) by developing self-supervised machine learning algorithms for classifying power system events using real-world Phasor Measurement Unit (PMU) data.
- Addressed challenges associated with noisy, incomplete measurements and validated models for use in smart grid applications.

Peer Reviewer

2023–Present

Regular reviewer for leading journals and conferences, including:

- IEEE Internet of Things Journal
- IEEE Journal of the Communications Society
- IEEE Transactions on Wireless Communications
- IEEE/ACM Transactions on Networking
- IEEE Transactions on Network Science and Engineering
- Nature Communications
- ACM SIGMOD Conference
- Journal of Biomedical Informatics

Skills

Key Skills: Machine Learning • Deep Learning • Mathematical Analysis & Optimization

- Tools, and coding: Python, MATLAB, Simulink, Computer Networks, AutoCAD, DCIM, Git, LaTeX, Linux
- Frameworks and packages: PyTorch, TensorFlow, NumPy, Pandas, Flower
- Machine learning: Supervised and Unsupervised Learning, Reinforcement Learning, Distributed Learning, Regression, Tree-based Models (XGBoost), Optimization and Statistics, Data Structures, Sorting and Searching Algorithms, Graph Algorithms (DFS, BFS), Algorithm Development